

Script SQL con consultas CRUD y consultas JOIN funcionales sobre la base de datos existente.

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1. Insertar propietario

The screenshot shows the pgAdmin 4 interface with a SQL query window titled 'gestion_alojamientos/postgres@PostgreSQL 18'. The query contains two main parts: a comment '-- 20 CONSULTAS SQL' and an SQL statement to insert a proprietor. The SQL statement is:

```
-- 01. INSERT - Insertar propietario
INSERT INTO propietarios (nombre, apellido, email, telefono)
VALUES ('Roberto', 'Hernández', 'roberto.h@email.com', '+503-7890-1234');
SELECT * FROM propietarios WHERE email='roberto.h@email.com';
```

The Data Output pane shows the result of the query, displaying a single row for the proprietor:

id_propietario	nombre	apellido	email	telefono	fecha_registro
6	Roberto	Hernández	roberto.h@email.com	+503-7890-1234	2026-05-30

2. Insertar alojamiento

The screenshot shows the pgAdmin 4 interface with a SQL query window titled 'consultas_sql.sql'. The query contains an SQL statement to insert a accommodation:

```
-- 02. INSERT - Insertar alojamiento vinculado
INSERT INTO alojamientos (id_propietario, nombre, tipo, direccion, ciudad, pais, precio_noche)
VALUES (6, 'Apartamento Vista Hermosa', 'apartamento', 'Av. Magnolias #456', 'San Salvador', 'El Salvador', 10);
RETURNING *;
```

The Data Output pane shows the result of the query, displaying a single row for the accommodation:

id_alojamiento	id_propietario	nombre	descripcion	tipo	direccion	ciudad	pais	precio_noche
11	6	Apartamento Vista Hermosa	[null]	apartamento	Av. Magnolias #456	San Salvador	El Salvador	10

3. Huésped y reserva

The screenshot shows the pgAdmin 4 interface with a SQL query window titled 'consultas_sql.sql'. The query contains an SQL statement to insert a guest:

```
-- 03. INSERT - Huésped y reserva
INSERT INTO huéspedes (nombre, apellido, email, nacionalidad)
VALUES ('Carlos', 'Gutiérrez', 'carlos.g@email.com', 'Costa Rica');
RETURNING *;
```

The Data Output pane shows the result of the query, displaying a single row for the guest:

id_huésped	nombre	apellido	email	telefono	nacionalidad	fecha_registro
11	Carlos	Gutiérrez	carlos.g@email.com	[null]	Costa Rica	2026-05-30

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including tables like 'alojamientos', 'huespedes', 'pagos', 'propietarios', and 'resenas'. The main window shows a query window titled 'consultas_sql.sql' with the following SQL statement:

```

INSERT INTO reservas (id_alojamiento, id_huesped, fecha_entrada, fecha_salida, num_personas, precio_total)
VALUES (11, 11, '2026-06-10', '2026-06-15', 3, 425.00)
RETURNING *;

```

The Data Output pane shows the result of the query:

	id_reserva [PK] integer	id_alojamiento integer	id_huesped integer	fecha_entrada date	fecha_salida date	num_personas integer	precio_total numeric (10,2)	estado character varying (50)	fecha_reserva timestamp without time zone
1	15	11	11	2026-06-10	2026-06-15	3	425.00	confirmada	2026-05-30 22:26:20.083893

4. Insertar pago

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure. The main window shows a query window titled 'consultas_sql.sql' with the following SQL statement:

```

-- 04. INSERT - Insertar pago
INSERT INTO pagos (id_reserva, monto, metodo_pago)
VALUES (15, 425.00, 'tarjeta')
RETURNING *;

```

The Data Output pane shows the result of the query:

	id_pago [PK] integer	id_reserva integer	monto numeric (10,2)	metodo_pago character varying (50)	estado_pago character varying (50)	fecha_pago timestamp without time zone
1	14	15	425.00	tarjeta	completado	2026-05-30 22:26:43.793956

5. Alojamientos activos

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure. The main window shows a query window titled 'consultas_sql.sql' with the following SQL statement:

```

-- 05. SELECT - Alojamientos activos
SELECT nombre, tipo, ciudad, precio_noche
FROM alojamientos
WHERE activo = true;

```

The Data Output pane shows the result of the query:

	nombre character varying (200)	tipo character varying (50)	ciudad character varying (100)	precio_noche numeric (10,2)
1	Apartamento Centro Históric...	apartamento	San Salvador	45.00
2	Casa de Playa El Tunco	casa	La Libertad	120.00
3	Villa Lago de Coatepeque	villa	Santa Ana	200.00
4	Habitación Privada San Mig...	habitacion	San Miguel	75.00
5	Apartamento Zona Rosa	apartamento	San Salvador	65.00
6	Casa Ruta de las Flores	casa	Sonsonate	80.00
7	Cabaña Montaña Cerro Verde	casa	Santa Ana	55.00
8	Loft Moderno Santa Tecla	apartamento	La Libertad	75.00
9	Penthouse San Benito	apartamento	San Salvador	150.00
10	Apartamento Vista Hermosa	apartamento	San Salvador	85.00

6. Huéspedes por país

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including tables like 'huespedes'. The main window shows a SQL query in the 'consultas_sql.sql' editor:

```
-- 06. SELECT - Huéspedes por país
SELECT nombre, apellido, email, nacionalidad
FROM huespedes
WHERE nacionalidad = 'Estados Unidos';
```

The 'Data Output' tab shows the results of the query:

	nombre	apellido	email	nacionalidad
1	Laura	Smith	laura.smith@email.com	Estados Unidos
2	Emma	Johnson	emma.johnson@email.c...	Estados Unidos

7. Reservas por fechas

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including tables like 'reservas'. The main window shows a SQL query in the 'consultas_sql.sql' editor:

```
-- 07. SELECT - Reservas por fechas (BETWEEN)
SELECT id_reserva, fecha_entrada, fecha_salida, precio_total
FROM reservas
WHERE fecha_entrada BETWEEN '2025-07-01' AND '2025-08-31';
```

The 'Data Output' tab shows the results of the query:

	id_reserva	fecha_entrada	fecha_salida	precio_total
1	2	2025-07-01	2025-07-07	720.00
2	3	2025-08-10	2025-08-15	1000.00
3	5	2025-07-15	2025-07-18	195.00
4	6	2025-08-01	2025-08-05	320.00
5	8	2025-07-20	2025-07-25	275.00
6	9	2025-08-15	2025-08-18	225.00

8. Actualizar precio

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including tables like 'alojamientos'. The main window shows a SQL query in the 'consultas_sql.sql' editor:

```
-- 08. UPDATE - Actualizar precio
UPDATE alojamientos
SET precio_noche = 50.00
WHERE id_alojamiento = 1
returning *;
```

The 'Data Output' tab shows the results of the query:

	id_alojamiento	id_propietario	nombre	descripcion	tipo	direccion	ciudad
1	1	1	Apartamento Centro Histó...	Hermoso apartamento en el corazón de San Salva...	apartamento	Calle Arce #245	San Salvador

9. Estado reserva

The screenshot shows the pgAdmin 4 interface with a query editor. The query is as follows:

```
-- 09. UPDATE - Estado reserva
UPDATE reservas
SET estado = 'completada'
WHERE id_reserva = 5
RETURNING *;
```

The Data Output tab shows the result of the query:

id_reserva	id_alojamiento	id_huesped	fecha_entrada	fecha_salida	num_personas	precio_total	estado	fecha_reserva
5	5	5	2025-07-15	2025-07-18	3	195.00	completada	2026-05-30 22:14:41.07435

10. Eliminar reseña

The screenshot shows the pgAdmin 4 interface with a query editor. The query is as follows:

```
-- 10. DELETE - Eliminar reseña
DELETE FROM resenas
WHERE id_resena = 3;
```

The Data Output tab shows the result of the query:

DELETE 1
Query returned successfully in 93 msec.

11. Reservas + huésped

The screenshot shows the pgAdmin 4 interface with a query editor. The query is as follows:

```
-- 11. JOIN - Reservas + huésped (INNER JOIN)
SELECT
  r.id_reserva,
  r.fecha_entrada,
  r.precio_total,
  h.nombre,
  h.apellido,
  h.nacionalidad
FROM reservas r
INNER JOIN huéspedes h ON r.id_huesped = h.id_huesped;
```

The Data Output tab shows the result of the query:

id_reserva	fecha_entrada	precio_total	nombre	apellido	nacionalidad
1	2025-06-15	225.00	Laura	Smith	Estados Unidos
2	2025-07-01	720.00	Juan	Pérez	El Salvador
3	2025-08-10	1000.00	Sophie	Dubois	Francia
4	2025-06-20	125.00	Marco	Rossi	Italia
5	2025-08-01	320.00	Diego	Fernández	México
6	2025-09-10	225.00	Yuki	Tanaka	Japón
7	2025-07-20	275.00	Hans	Mueller	Alemania
8	2025-08-15	225.00	Isabella	Costa	Brasil
9	2025-09-01	600.00	Oliver	Brown	Reino Unido
10	2025-10-01	720.00	Laura	Smith	Estados Unidos
11	2025-11-15	1000.00	Emma	Johnson	Estados Unidos

Successfully run. Total query runtime: 108 msec. 15 rows affected.

12. Alojamiento completo

The screenshot shows the pgAdmin 4 interface with a SQL query executed in the 'consultas_sql.sql*' window. The query is a complex JOIN involving 'alojamientos', 'propietarios', and 'reservas' tables. The results are displayed in the 'Data Output' pane, showing 15 rows of data with columns: alojamiento, ciudad, propietario, and fecha_entrada.

```
-- 12. JOIN - Alojamiento completo (INNER JOIN múltiple)
SELECT
  a.nombre AS alojamiento,
  a.ciudad,
  p.nombre AS propietario,
  r.fecha_entrada
FROM alojamientos a
INNER JOIN propietarios p ON a.id_propietario = p.id_propietario
INNER JOIN reservas r ON a.id_alojamiento = r.id_alojamiento;
```

alojamiento	ciudad	propietario	fecha_entrada
Apartamento Centro Histó...	San Salvador	Carlos	2025-06-15
Casa de Playa El Tunco	La Libertad	Carlos	2025-07-01
Villa Lago de Coatepeque	Santa Ana	María	2025-08-10
Habitación Privada San Mig...	San Miguel	María	2025-06-20
Casa Ruta de las Flores	Sonsonate	José	2025-08-01
Apartamento Centro Histó...	San Salvador	Carlos	2025-09-10
Cabaña Montaña Cerro Verde	Santa Ana	Ana	2025-07-20
Loft Moderno Santa Tecla	La Libertad	Pedro	2025-08-15
Penthouse San Benito	San Salvador	Pedro	2025-09-01
Casa de Playa El Tunco	La Libertad	Carlos	2025-10-01
Villa Lago de Coatepeque	Santa Ana	María	2025-11-15

Successfully run. Total query runtime: 129 msec. 15 rows affected.

13. Pagos + reservas

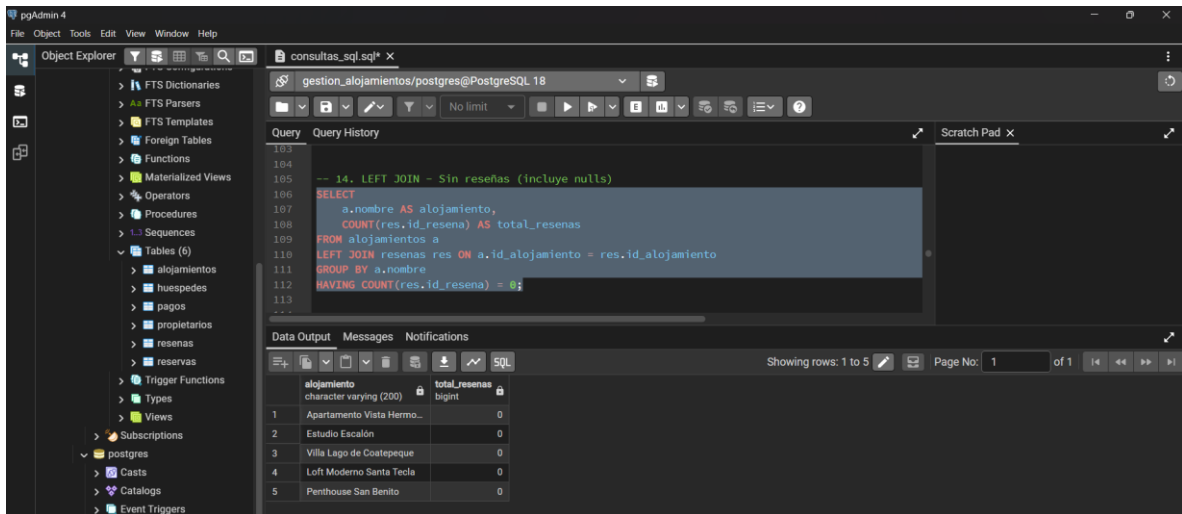
The screenshot shows the pgAdmin 4 interface with a SQL query executed in the 'consultas_sql.sql*' window. The query is a complex JOIN involving 'pagos', 'reservas', and 'huespedes' tables. The results are displayed in the 'Data Output' pane, showing 14 rows of data with columns: monto, metodo_pago, fecha_entrada, and huesped.

```
-- 13. JOIN - Pagos + reservas (JOIN combinado)
SELECT
  pag.monto,
  pag.metodo_pago,
  r.fecha_entrada,
  h.nombre AS huesped
FROM pagos pag
INNER JOIN reservas r ON pag.id_reserva = r.id_reserva
INNER JOIN huespedes h ON r.id_huesped = h.id_huesped;
```

monto	metodo_pago	fecha_entrada	huesped
225.00	tarjeta	2025-06-15	Laura
720.00	paypal	2025-07-01	Juan
1000.00	transferencia	2025-08-10	Sophie
125.00	tarjeta	2025-06-20	Marco
195.00	tarjeta	2025-07-15	Emma
320.00	paypal	2025-08-01	Diego
225.00	tarjeta	2025-09-10	Yuki
275.00	efectivo	2025-07-20	Hans
225.00	tarjeta	2025-08-15	Isabella
600.00	transferencia	2025-09-01	Oliver
720.00	tarjeta	2025-10-01	Laura

Successfully run. Total query runtime: 106 msec. 14 rows affected.

14. Sin reseñas



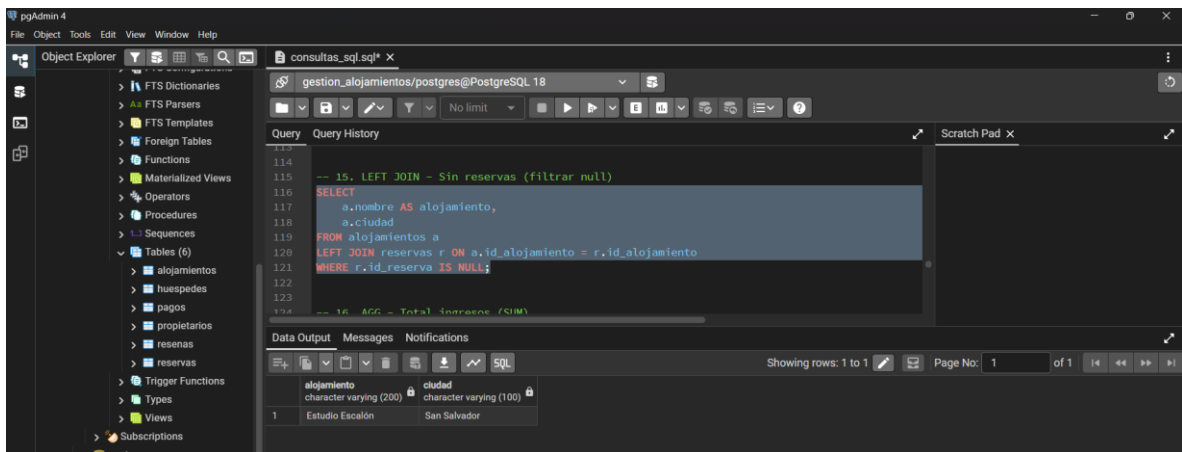
The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including tables like 'alojamientos', 'huespedes', 'pagos', 'propietarios', 'resenas', and 'reservas'. The main query editor shows the following SQL query:

```
-- 14. LEFT JOIN - Sin reseñas (incluye nulls)
SELECT
  a.nombre AS alojamiento,
  COUNT(res.id_resena) AS total_resenas
FROM alojamientos a
LEFT JOIN resenas res ON a.id_alojamiento = res.id_alojamiento
GROUP BY a.nombre
HAVING COUNT(res.id_resena) = 0;
```

The 'Data Output' tab shows the results of the query:

alojamiento	total_resenas
Apartamento Vista Hermosa	0
Estudio Escalón	0
Villa Lago de Coatepeque	0
Loft Moderno Santa Tecla	0
Penthouse San Benito	0

15. Sin reservas



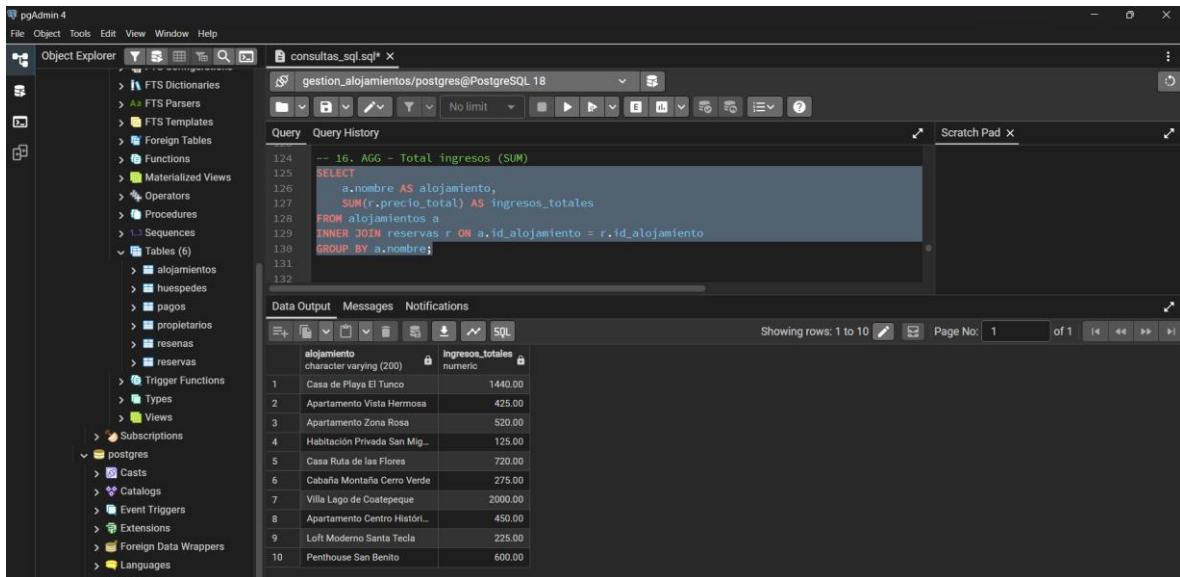
The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including tables like 'alojamientos', 'huespedes', 'pagos', 'propietarios', 'resenas', and 'reservas'. The main query editor shows the following SQL query:

```
-- 15. LEFT JOIN - Sin reservas (filtrar null)
SELECT
  a.nombre AS alojamiento,
  a.ciudad
FROM alojamientos a
LEFT JOIN reservas r ON a.id_alojamiento = r.id_alojamiento
WHERE r.id_reserva IS NULL;
```

The 'Data Output' tab shows the results of the query:

alojamiento	ciudad
Estudio Escalón	San Salvador

16. Total ingresos

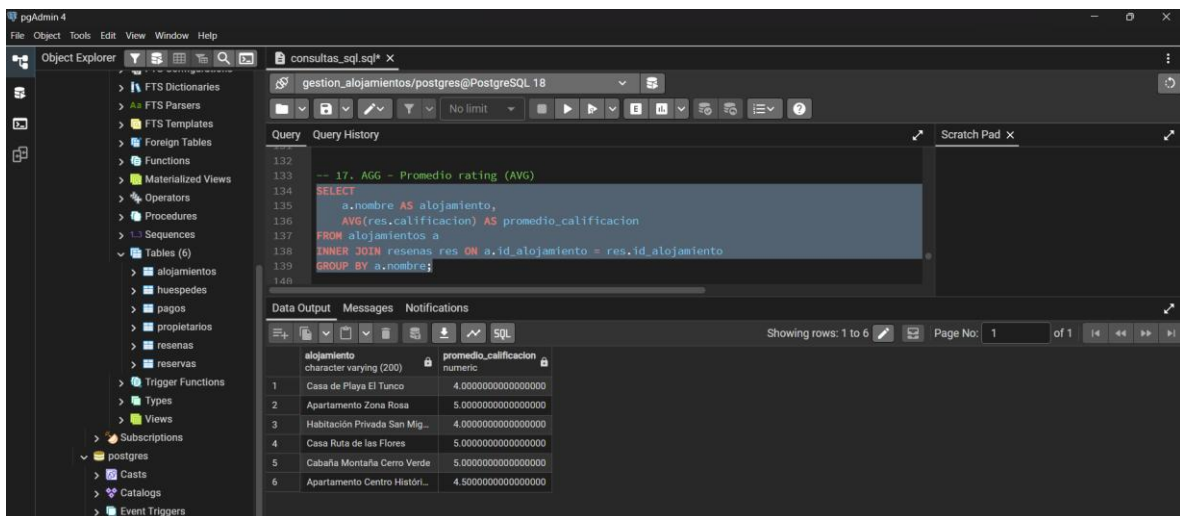


The screenshot shows the pgAdmin 4 interface with a SQL query executed in the 'consultas_sql.sql' file. The query calculates the total income for each accommodation type. The results are displayed in the 'Data Output' pane, showing 10 rows of data.

```
-- 16. AGG - Total ingresos (SUM)
SELECT
  a.nombre AS alojamiento,
  SUM(r.precio_total) AS ingresos_totales
FROM alojamientos a
INNER JOIN reservas r ON a.id_alojamiento = r.id_alojamiento
GROUP BY a.nombre;
```

alojamiento	ingresos_totales
1 Casa de Playa El Tunco	1440.00
2 Apartamento Vista Hermosa	425.00
3 Apartamento Zona Rosa	520.00
4 Habitación Privada San Mig...	125.00
5 Casa Ruta de las Flores	720.00
6 Cabaña Montaña Cerro Verde	275.00
7 Villa Lago de Coatepeque	2000.00
8 Apartamento Centro Históri...	450.00
9 Loft Moderno Santa Tecla	225.00
10 Penthouse San Benito	600.00

17. Promedio rating



The screenshot shows the pgAdmin 4 interface with a SQL query executed in the 'consultas_sql.sql' file. The query calculates the average rating for each accommodation type. The results are displayed in the 'Data Output' pane, showing 6 rows of data.

```
-- 17. AGG - Promedio rating (AVG)
SELECT
  a.nombre AS alojamiento,
  AVG(res.calificacion) AS promedio_calificacion
FROM alojamientos a
INNER JOIN resenas res ON a.id_alojamiento = res.id_alojamiento
GROUP BY a.nombre;
```

alojamiento	promedio_calificacion
1 Casa de Playa El Tunco	4.0000000000000000
2 Apartamento Zona Rosa	5.0000000000000000
3 Habitación Privada San Mig...	4.0000000000000000
4 Casa Ruta de las Flores	5.0000000000000000
5 Cabaña Montaña Cerro Verde	5.0000000000000000
6 Apartamento Centro Históri...	4.5000000000000000

18. Top alojamientos

The screenshot shows the pgAdmin 4 interface with a SQL query editor. The query is as follows:

```
-- 18. AGG - Top alojamientos (COUNT + LIMIT)
SELECT
  a.nombre AS alojamiento,
  COUNT(r.id_reserva) AS total_reservas
FROM alojamientos a
INNER JOIN reservas r ON a.id_alojamiento = r.id_alojamiento
GROUP BY a.nombre
ORDER BY total_reservas DESC
LIMIT 5;
```

The Data Output pane shows the results of the query:

alojamiento	total_reservas
1 Apartamento Zona Rosa	2
2 Villa Lago de Coatepeque	2
3 Apartamento Centro Histó...	2
4 Casa Ruta de las Flores	2
5 Casa de Playa El Tunco	2

19. Más de 3 reservas

The screenshot shows the pgAdmin 4 interface with a SQL query editor. The query is as follows:

```
-- 19. HAVING - Más de 3 reservas (GROUP BY + HAVING)
SELECT
  a.nombre AS alojamiento,
  COUNT(r.id_reserva) AS total_reservas
FROM alojamientos a
INNER JOIN reservas r ON a.id_alojamiento = r.id_alojamiento
GROUP BY a.nombre
HAVING COUNT(r.id_reserva) > 3;
```

The Data Output pane shows the results of the query:

alojamiento	total_reservas
1 Casa de Playa El Tunco	2
2 Apartamento Zona Rosa	2
3 Casa Ruta de las Flores	2
4 Villa Lago de Coatepeque	2
5 Apartamento Centro Histó...	2

20. Alojamiento más caro

The screenshot shows the pgAdmin 4 interface with a SQL query editor. The query is as follows:

```
-- 20. Subconsulta - Alojamiento más caro (Subquery)
SELECT nombre, precio_noche
FROM alojamientos
WHERE precio_noche = (SELECT MAX(precio_noche) FROM alojamientos);
```

The Data Output pane shows the results of the query:

nombre	precio_noche
1 Villa Lago de Coatepeq...	200.00